ANT 6973: DATA VISUALIZATION AND EXPLORATION

COLOR, SAVING OUTPUT, AND GEOSPATIAL DATA

TODAY'S TOPICS

- Using color effectively
- Saving your plots
- More about visualizing geospatial data
- Small Project 1

USING COLOR EFFECTIVELY

TYPES OF COLOR PALETTES

- There are three main types of color palettes:
  - Qualitative
  - Sequential
  - Diverging
- Each is appropriate for particular kinds of data.

QUALITATIVE COLOR PALETTES

- Use when color shows discrete categories that *don't have intrinsic order.*

- Limitations: unlikely to work well with more than ~12 categories (3–5 ideal)

ggplot2 default



`Example: scale_color_hue()`

QUALITATIVE COLOR PALETTES

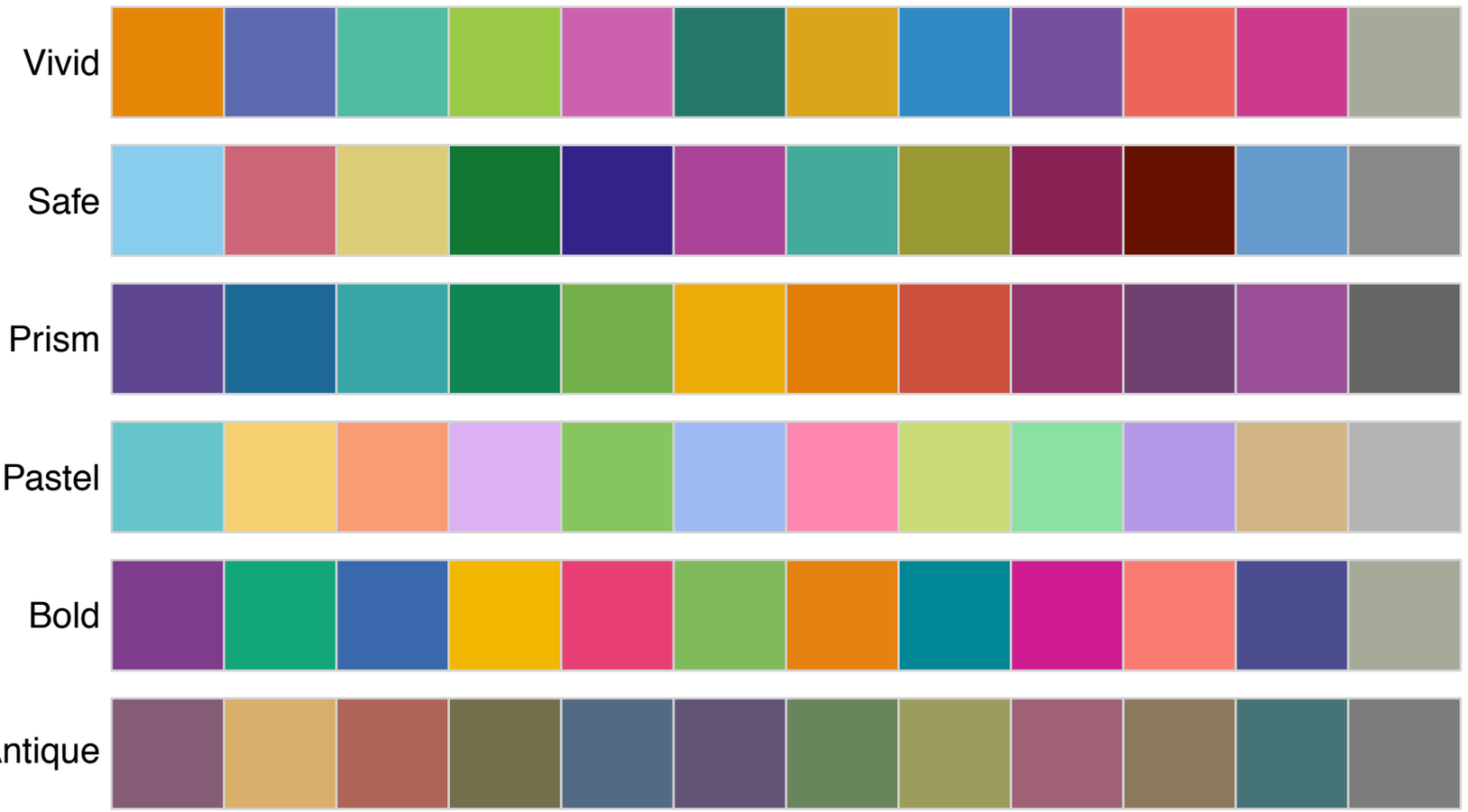
RColorBrewer




```
Example: scale_color_brewer(palette = "Set1")
```

QUALITATIVE COLOR PALETTES

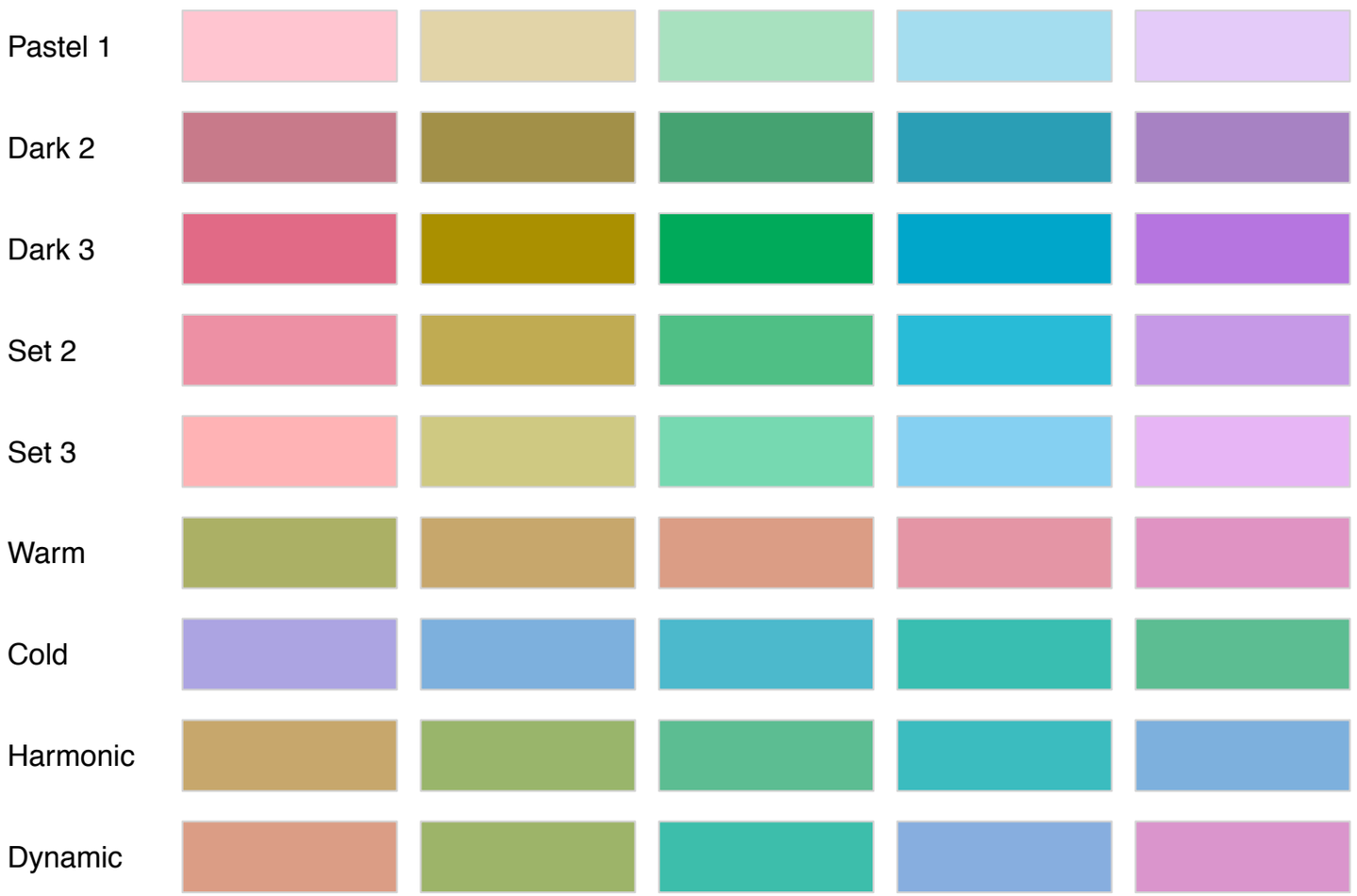
rcartocolor




```
Example: scale_color_carto_d(palette = "Bold")
```

QUALITATIVE COLOR PALETTES

colorspace




```
Example: scale_color_discrete_qualitative(palette = "Cold")
```

SEQUENTIAL COLOR PALETTES

- Use when color represents data values that have a natural order.

- Can be based on a single hue (e.g., from dark blue to light blue) or on multiple hues (e.g., from dark red to light yellow)

SEQUENTIAL COLOR PALETTES

- Two usage cases:

SEQUENTIAL COLOR PALETTES

- Two usage cases:
  - When you have ordered categories

ggplot2 default



`Example: scale_color_ordinal()`

SEQUENTIAL COLOR PALETTES

- Two usage cases:
  - When you have ordered categories
  - When you have continuously varying values

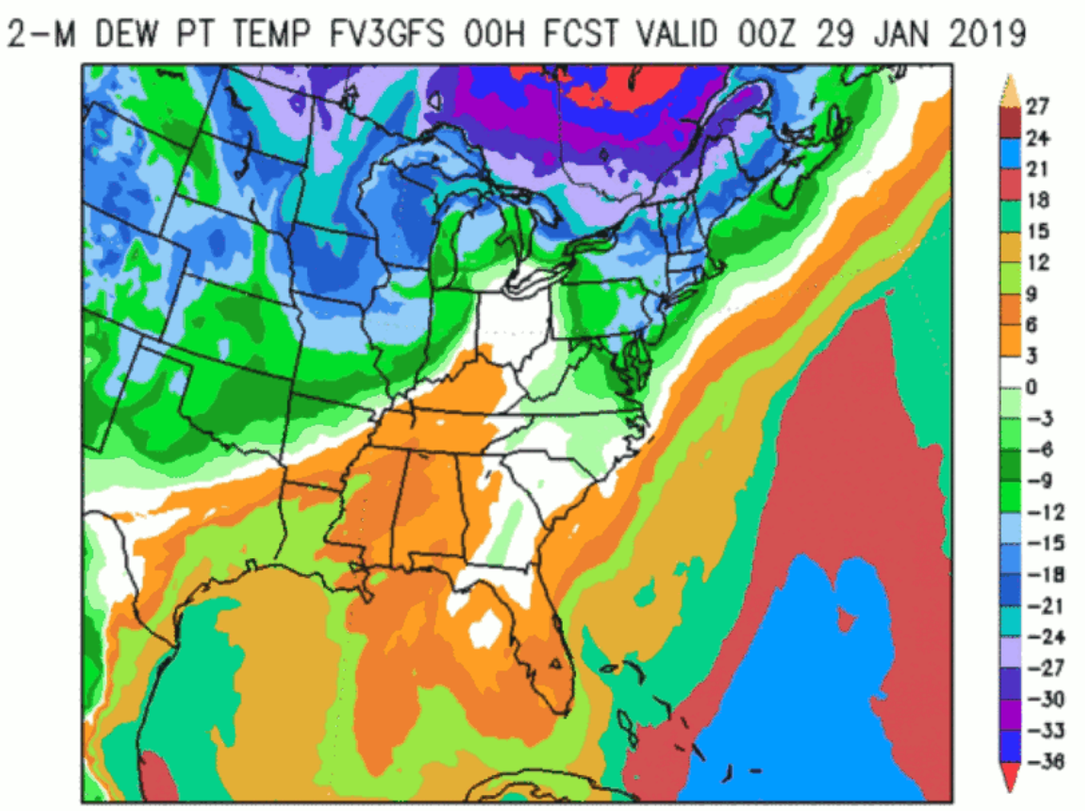
ggplot2 default

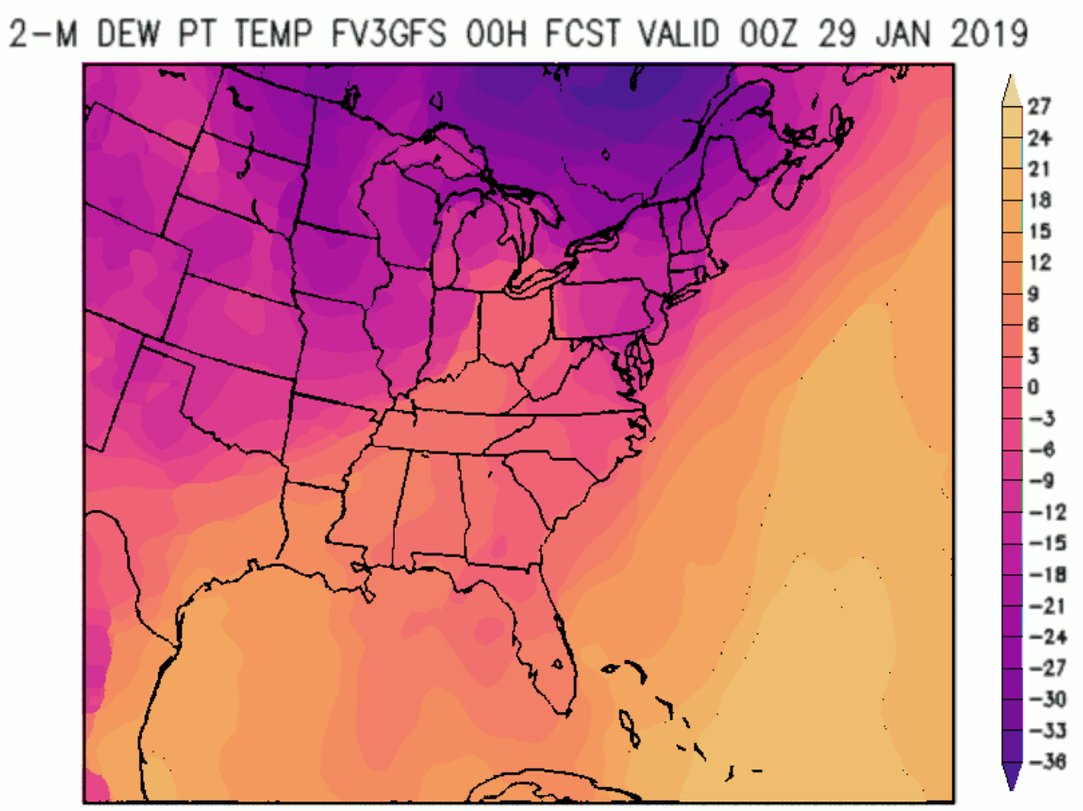


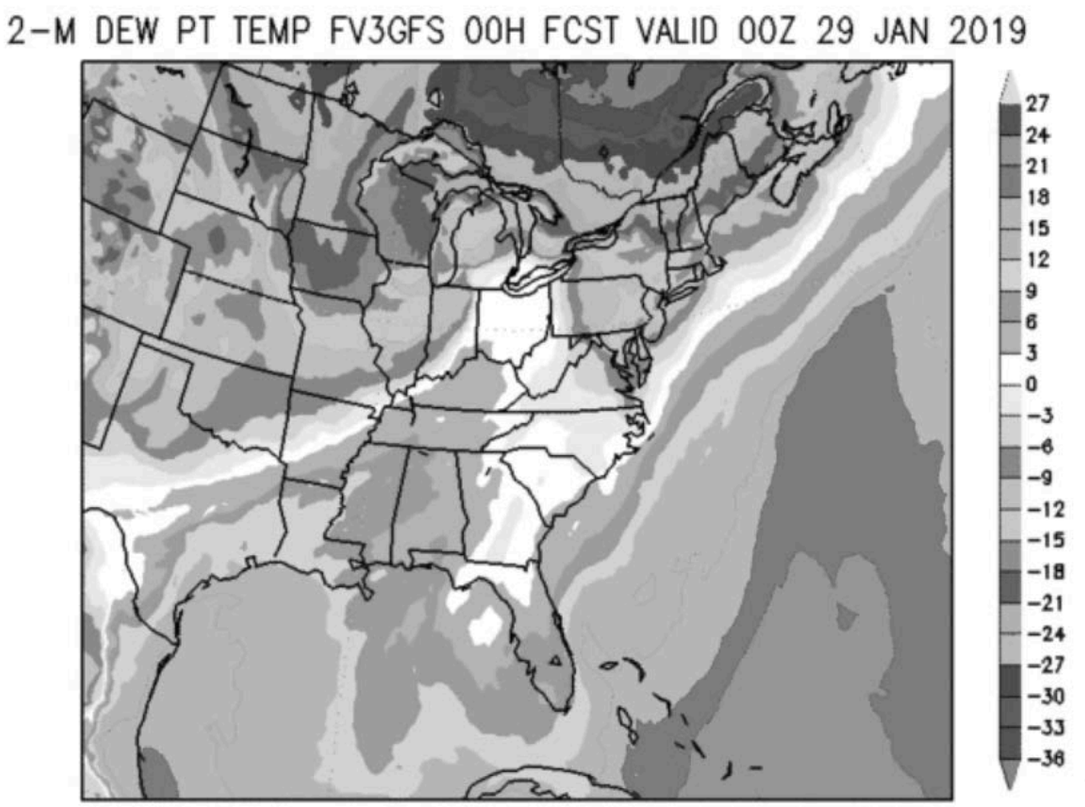
`Example: scale_color_continuous()`

SEQUENTIAL COLOR PALETTES

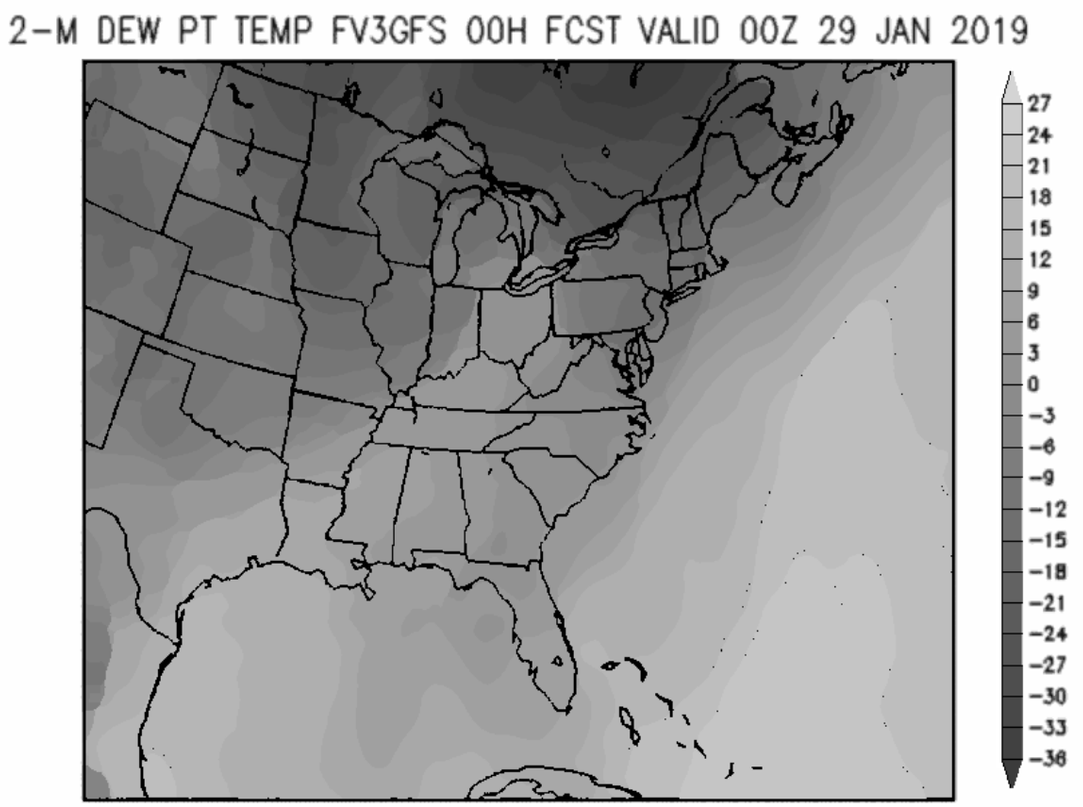
- A good sequential color scale accurately conveys:
  - Which values are larger or smaller
  - How distant two values are from each other
- This means that it should be *perceptually uniform* across its entire range.


2-M DEW PT TEMP FV3GFS 00H FCST VALID 00Z 29 JAN 2019
27
24
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18
15
12
9
6
3
0
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-12
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-27
-30
-33
-36


2-M DEW PT TEMP FV3GFS 00H FCST VALID 00Z 29 JAN 2019
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2-M DEW PT TEMP FV3GFS 00H FCST VALID 00Z 29 JAN 2019
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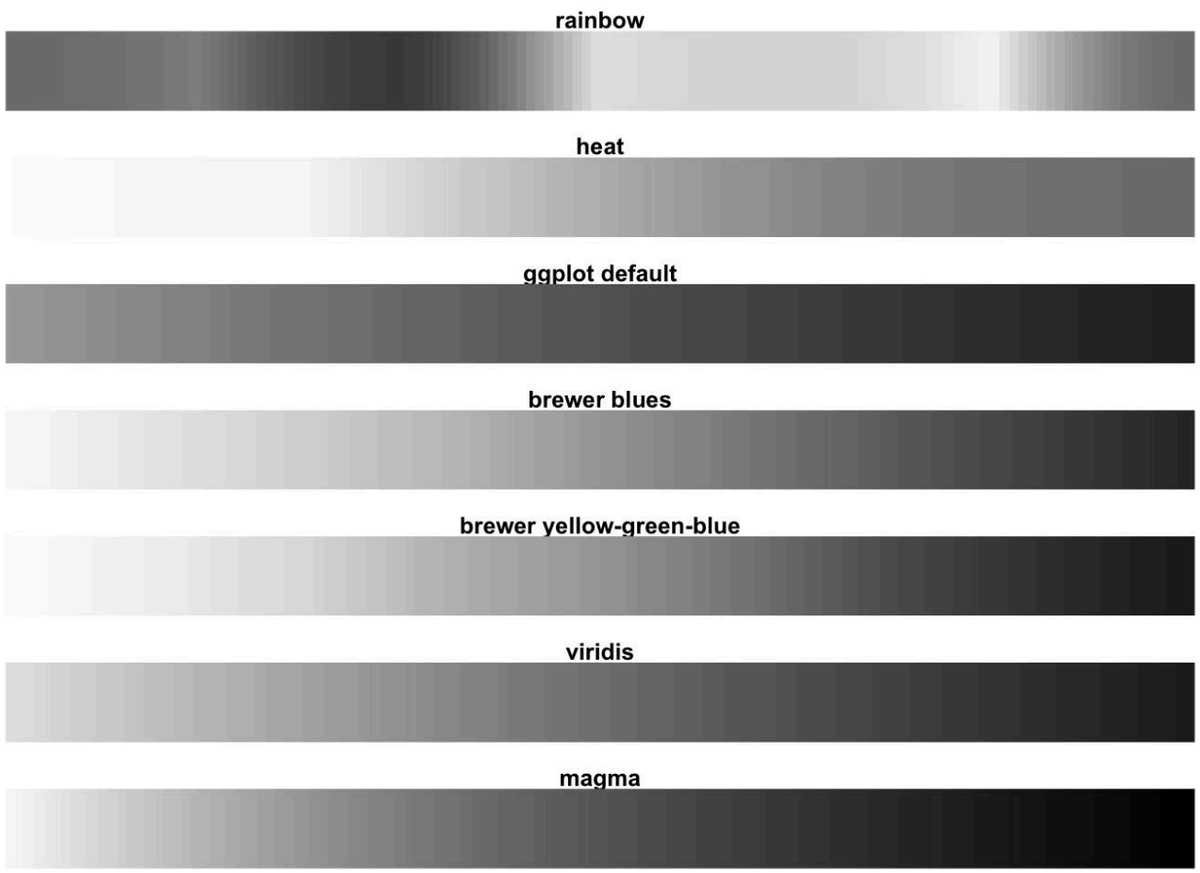
, ~1


2-M DEW PT TEMP FV3GFS 00H FCST VALID 00Z 29 JAN 2019
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-36

SEQUENTIAL COLOR PALETTES

Desaturated

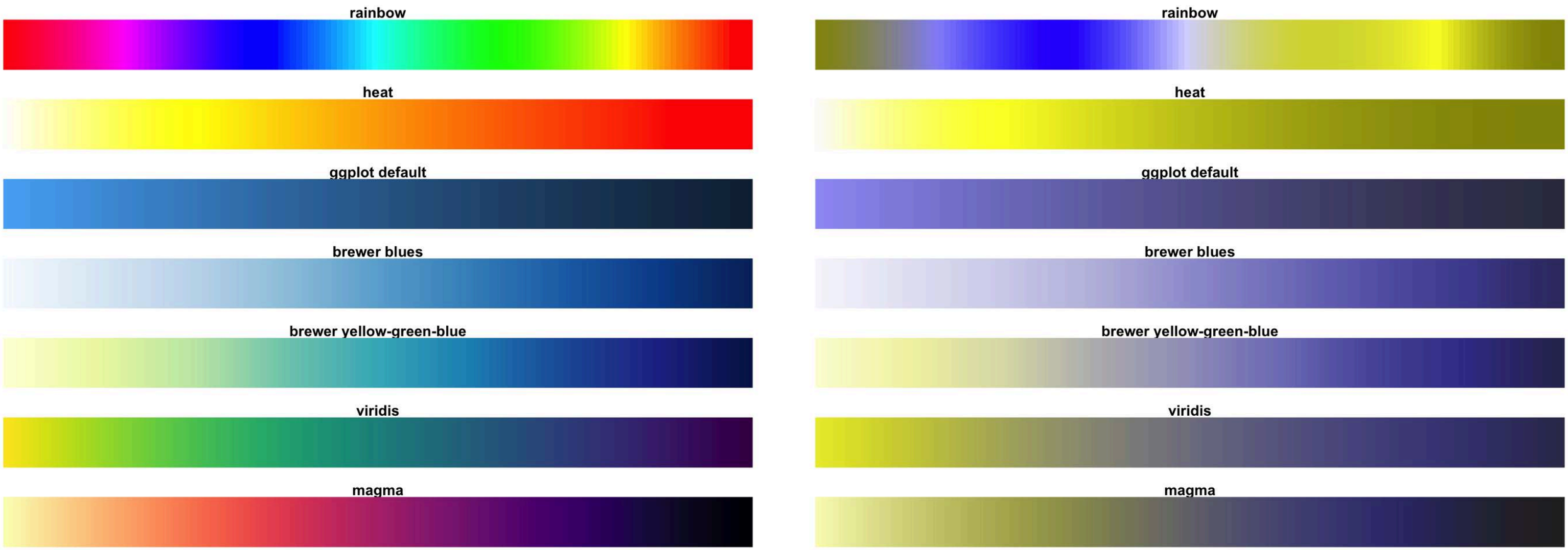




When figures are printed or rendered in grayscale

SEQUENTIAL COLOR PALETTES

Red-green color blindness

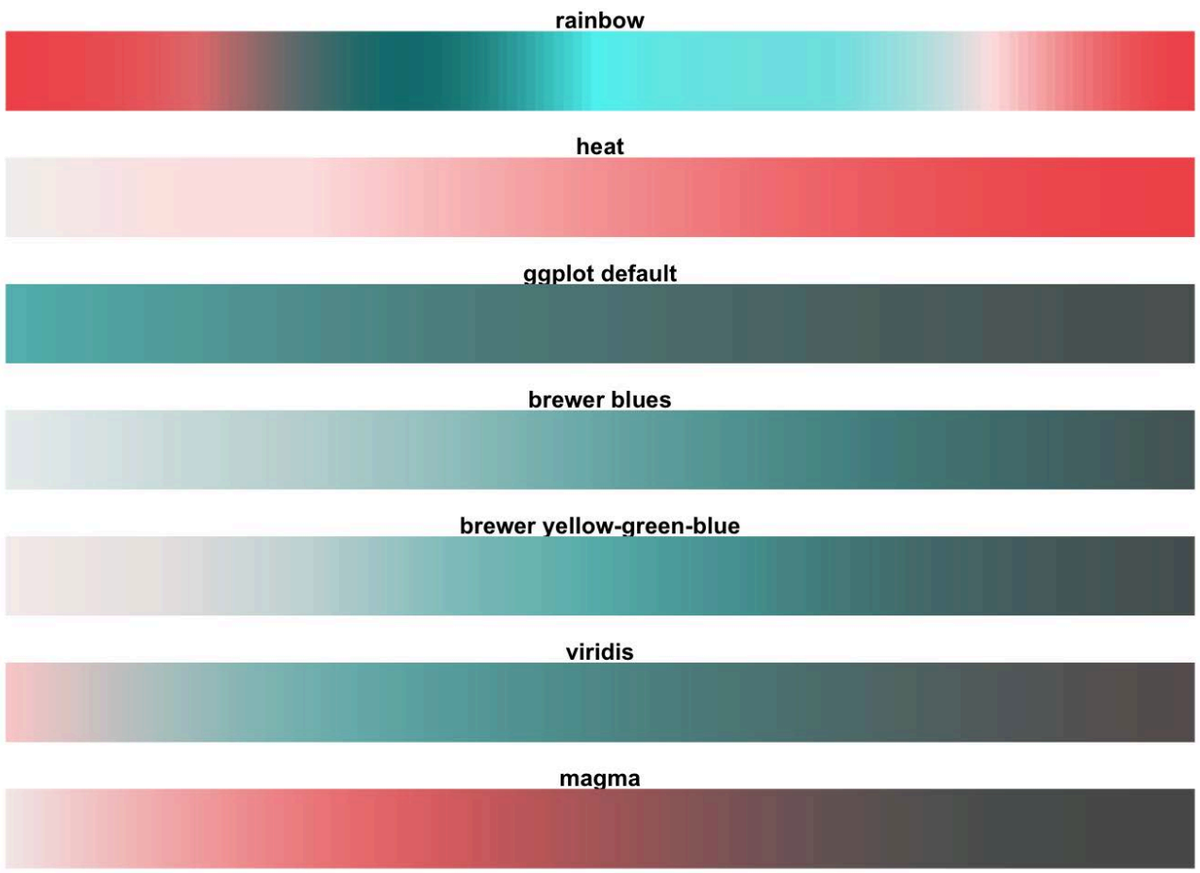


~8% of men and 0.5% of women

SEQUENTIAL COLOR PALETTES

Blue-yellow color blindness

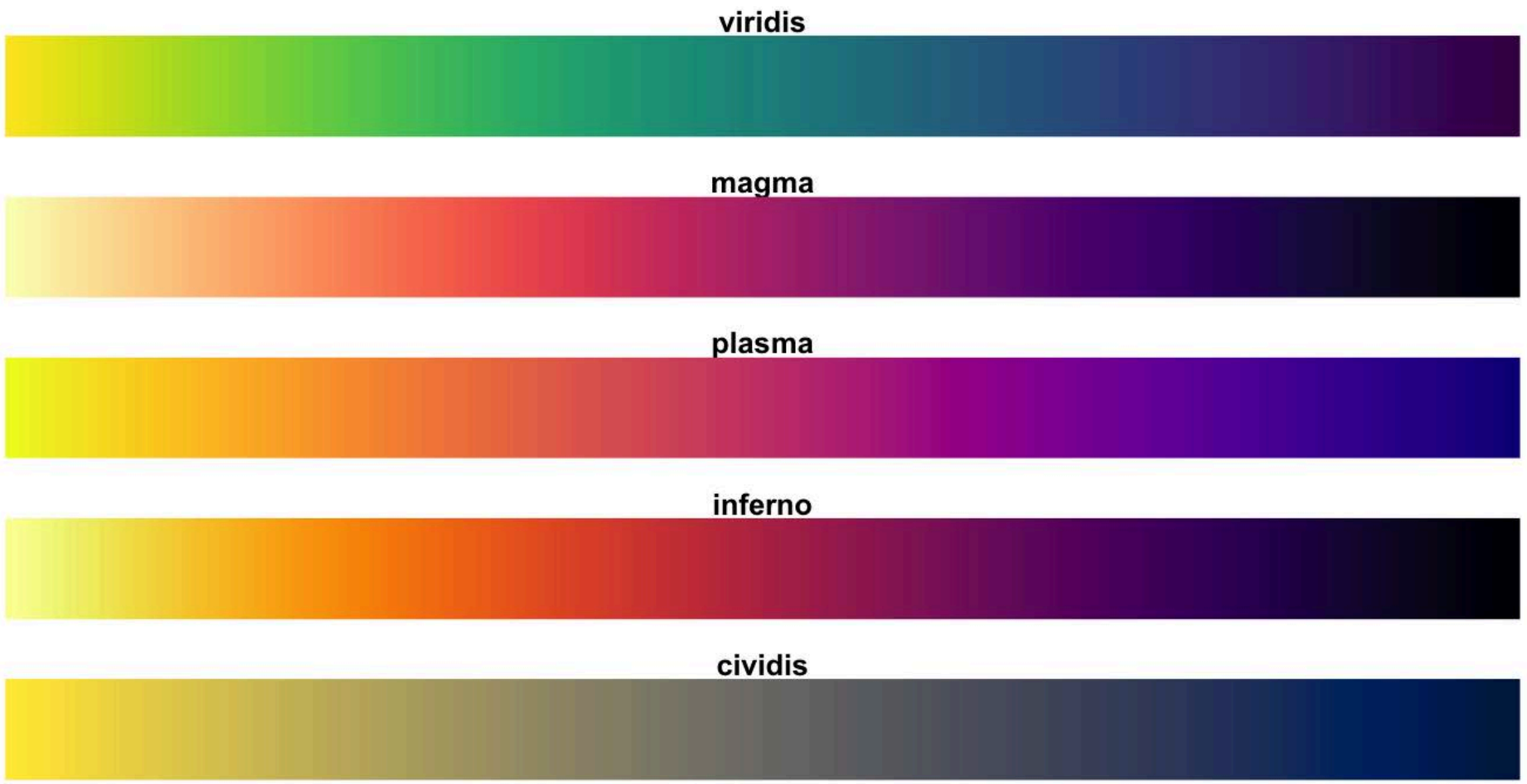




Very rare, ~1 in 40,000

SEQUENTIAL COLOR PALETTES

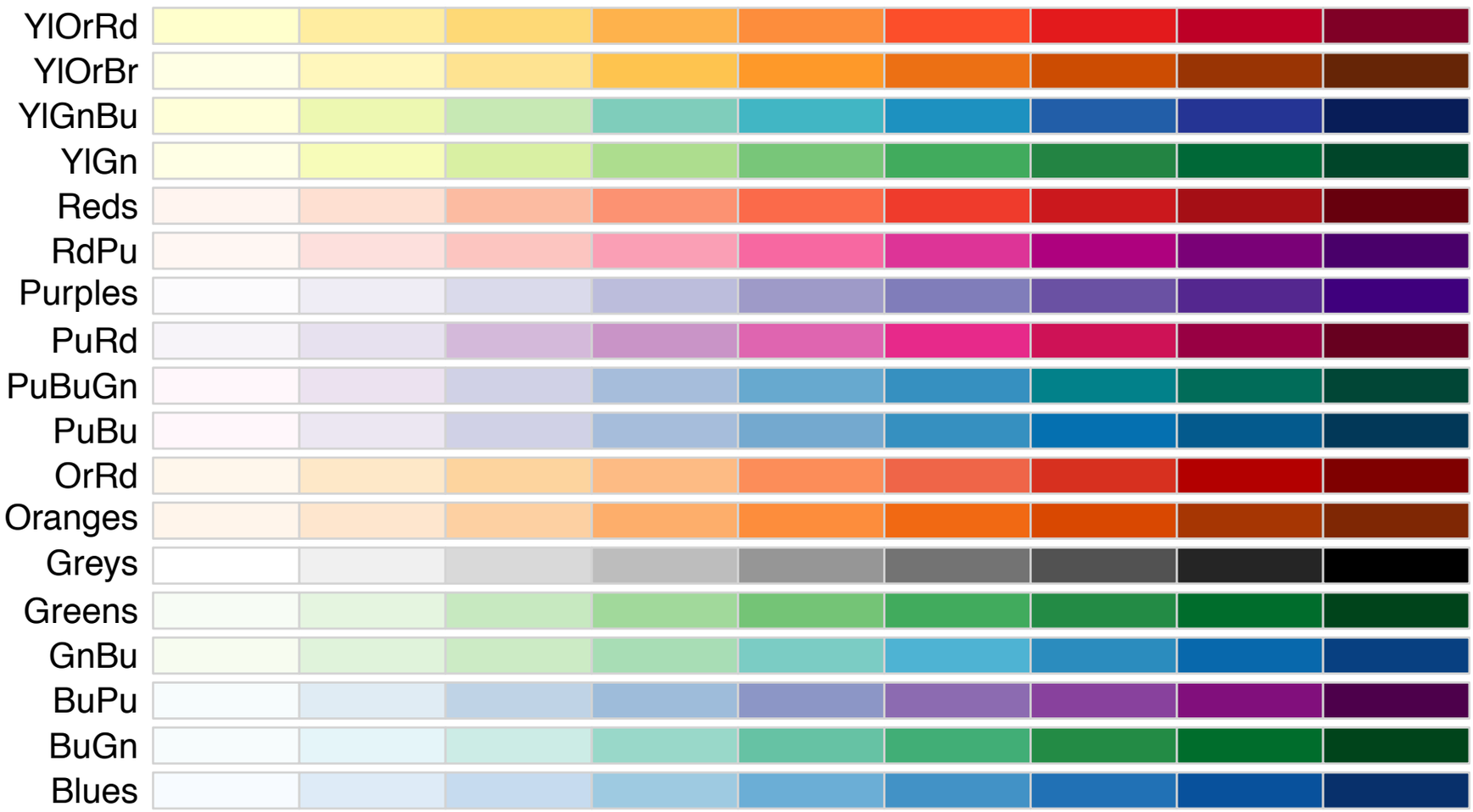
viridis




```
Ordered categories: scale_color_viridis_d(option = "magma")
Continuous values: scale_color_viridis_c(option = "magma")
```

SEQUENTIAL COLOR PALETTES

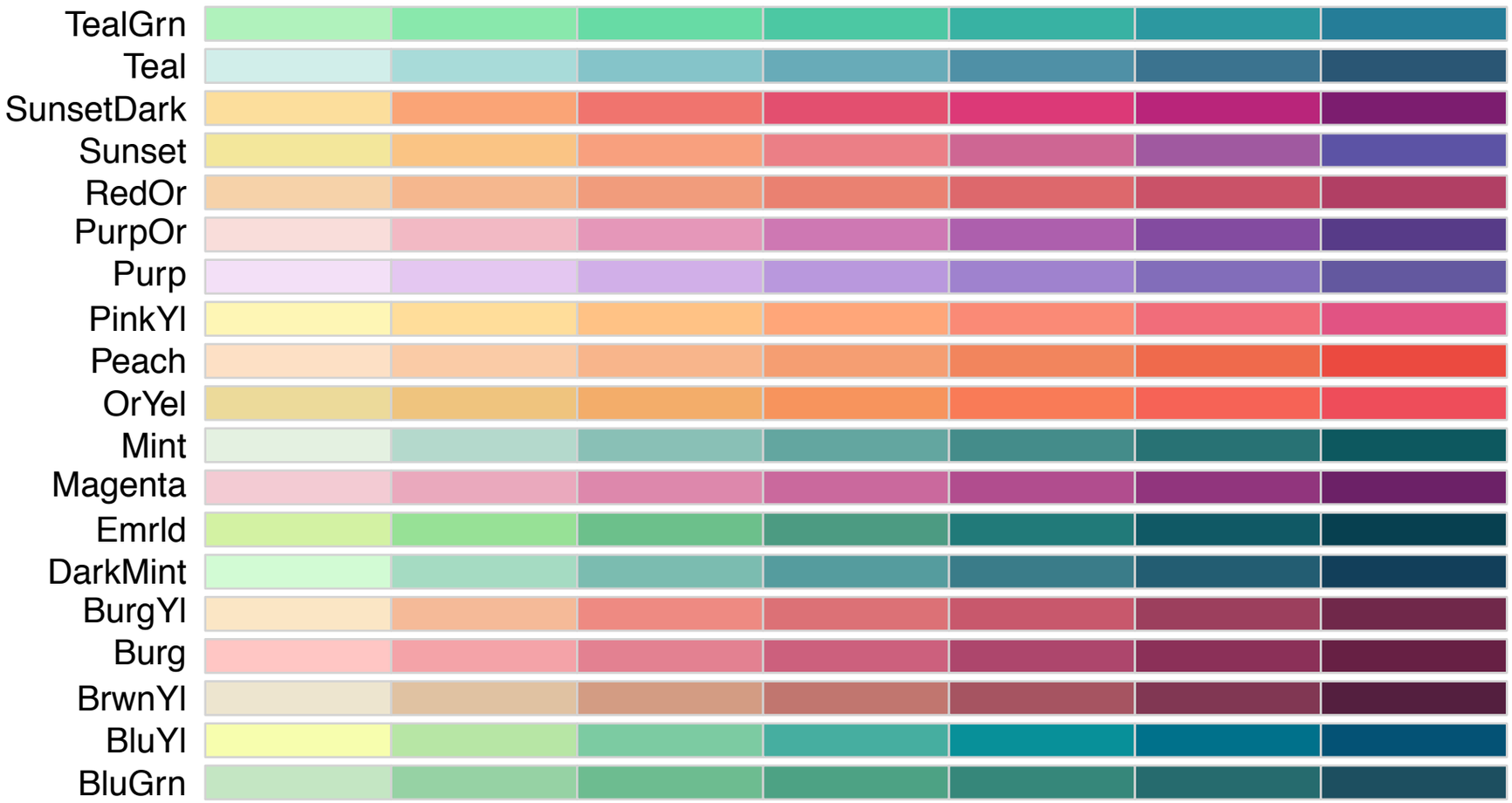
RColorBrewer




```
Ordered categories: scale_color_brewer(palette = "RdPu")
Continuous values: scale_color_distiller(palette = "RdPu")
```

SEQUENTIAL COLOR PALETTES

rcartocolor




```
Ordered categories: scale_color_carto_d(palette = "Mint")
Continuous values: scale_color_carto_c(palette = "Mint")
```

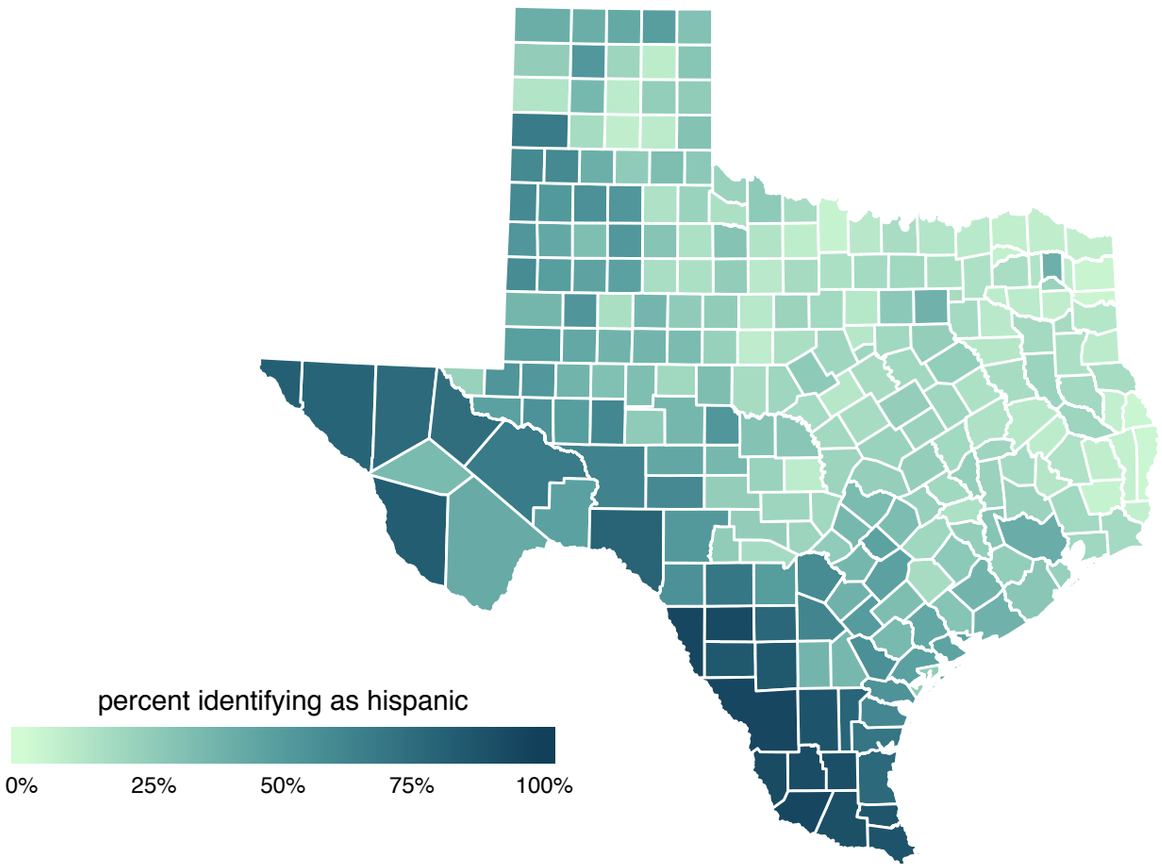
SEQUENTIAL COLOR PALETTES

colorspace

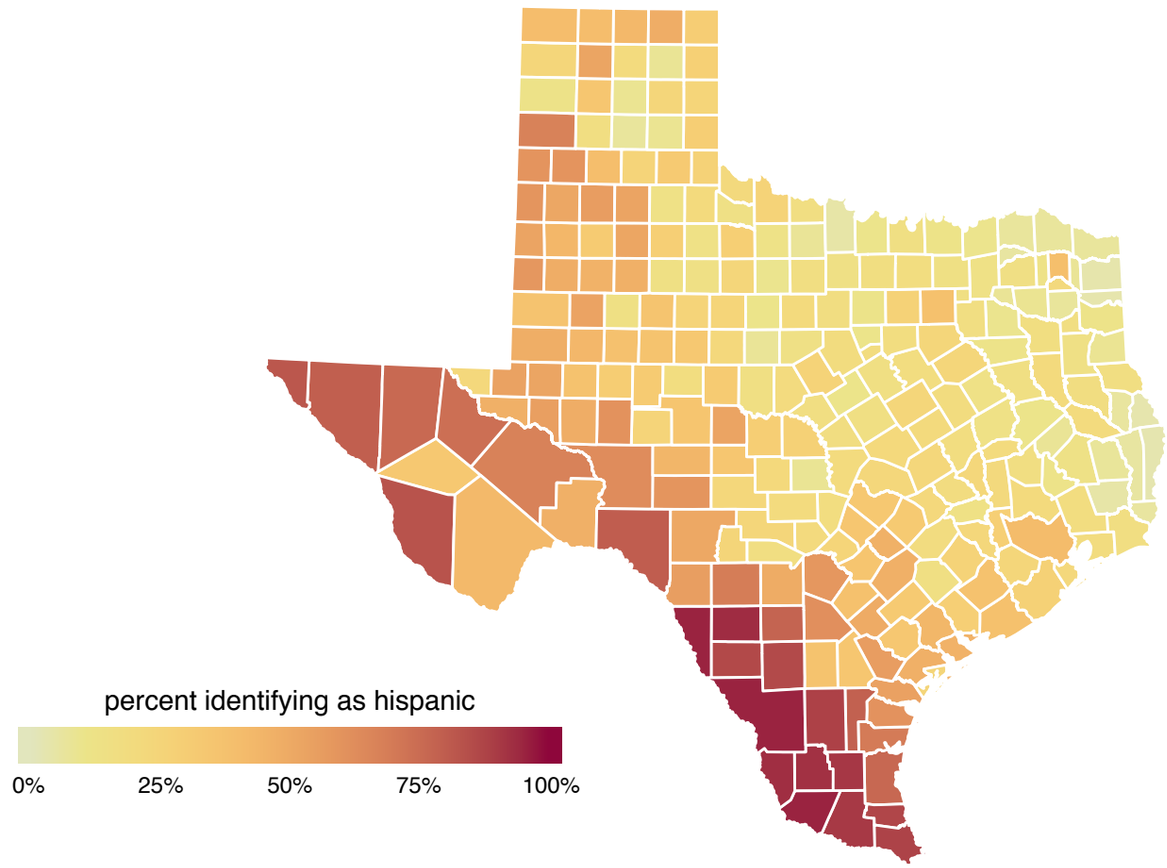



```
Ordered categories: scale_color_discrete_sequential(palette = "Heat")
Continuous values: scale_color_continuous_sequential(palette = "Heat")
```

SEQUENTIAL COLOR PALETTES



DarkMint from rcartocolor

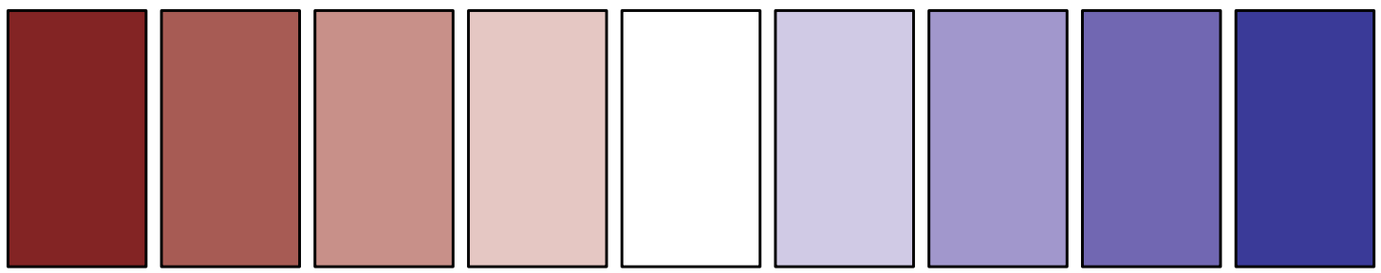


Heat from colorspace

DIVERGING COLOR PALETTES

- Use when data values diverge in two directions from a neutral midpoint (e.g., positive and negative numbers).
- Shows both how much and in which direction each value deviates from the midpoint.

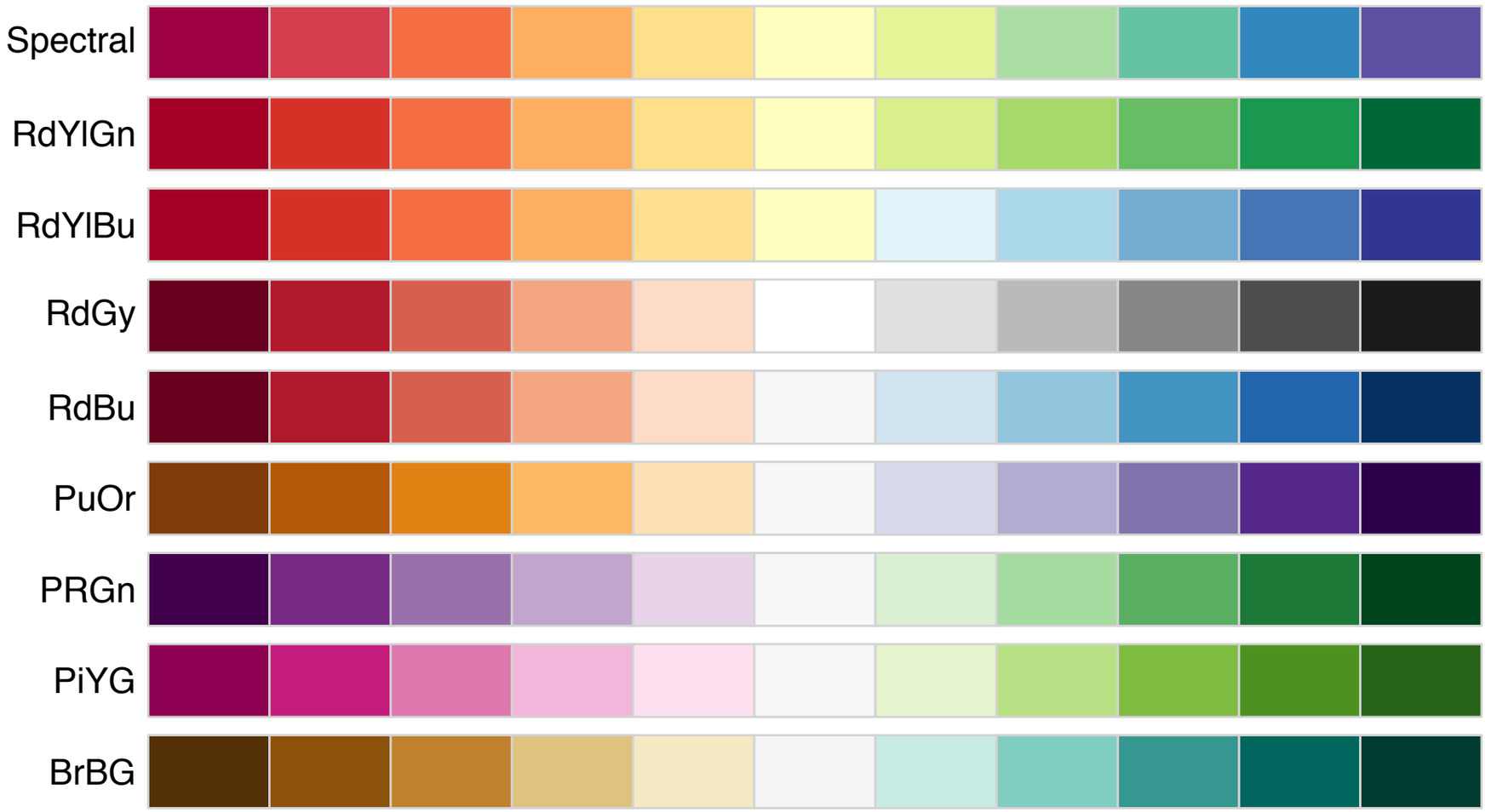
ggplot2 default



`Example: scale_color_gradient2()`

DIVERGING COLOR PALETTES

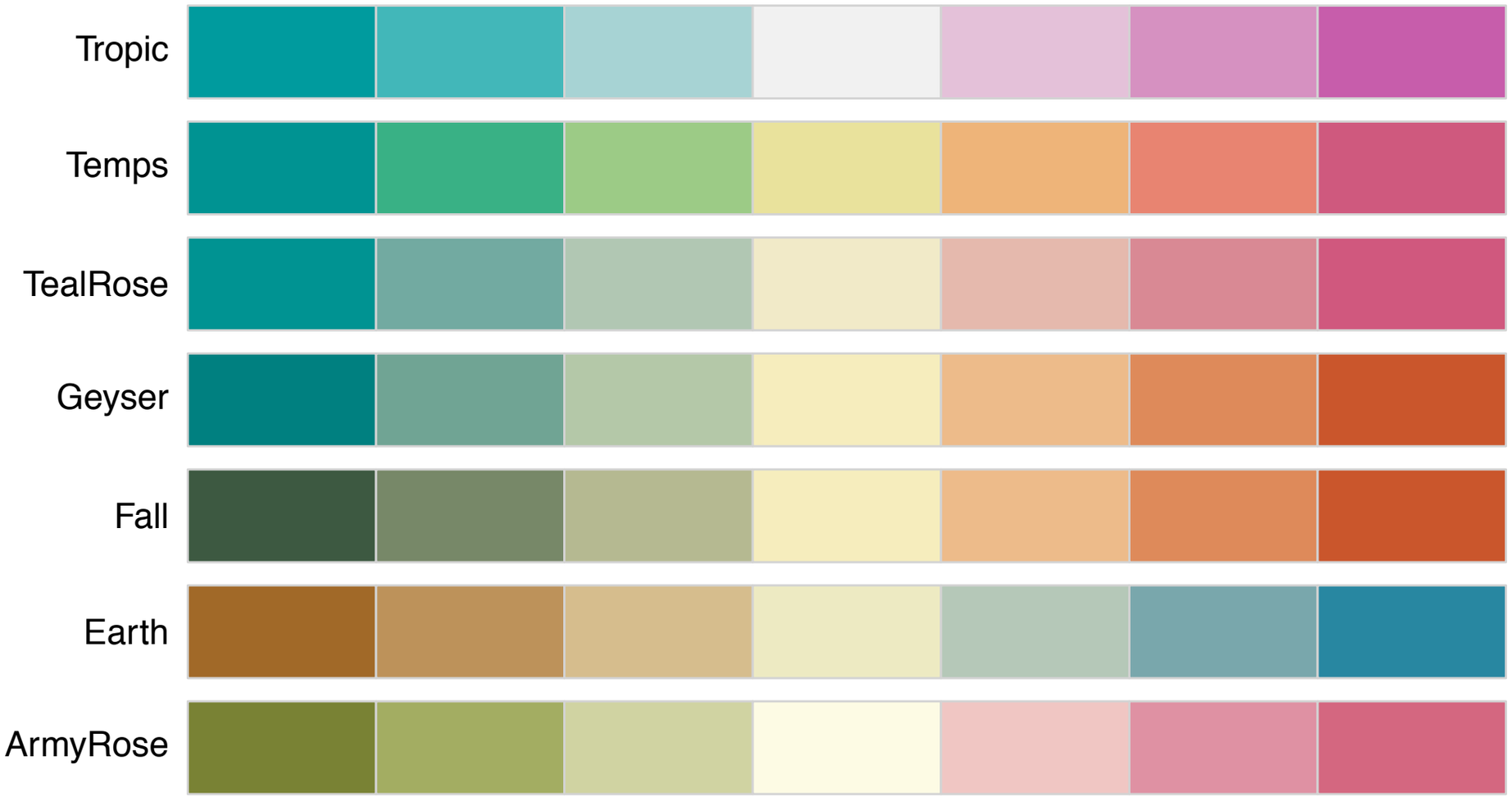
RColorBrewer




```
Ordered categories: scale_color_brewer(palette = "RdYlBu")
Continuous values: scale_color_distiller(palette = "RdYlBu")
```

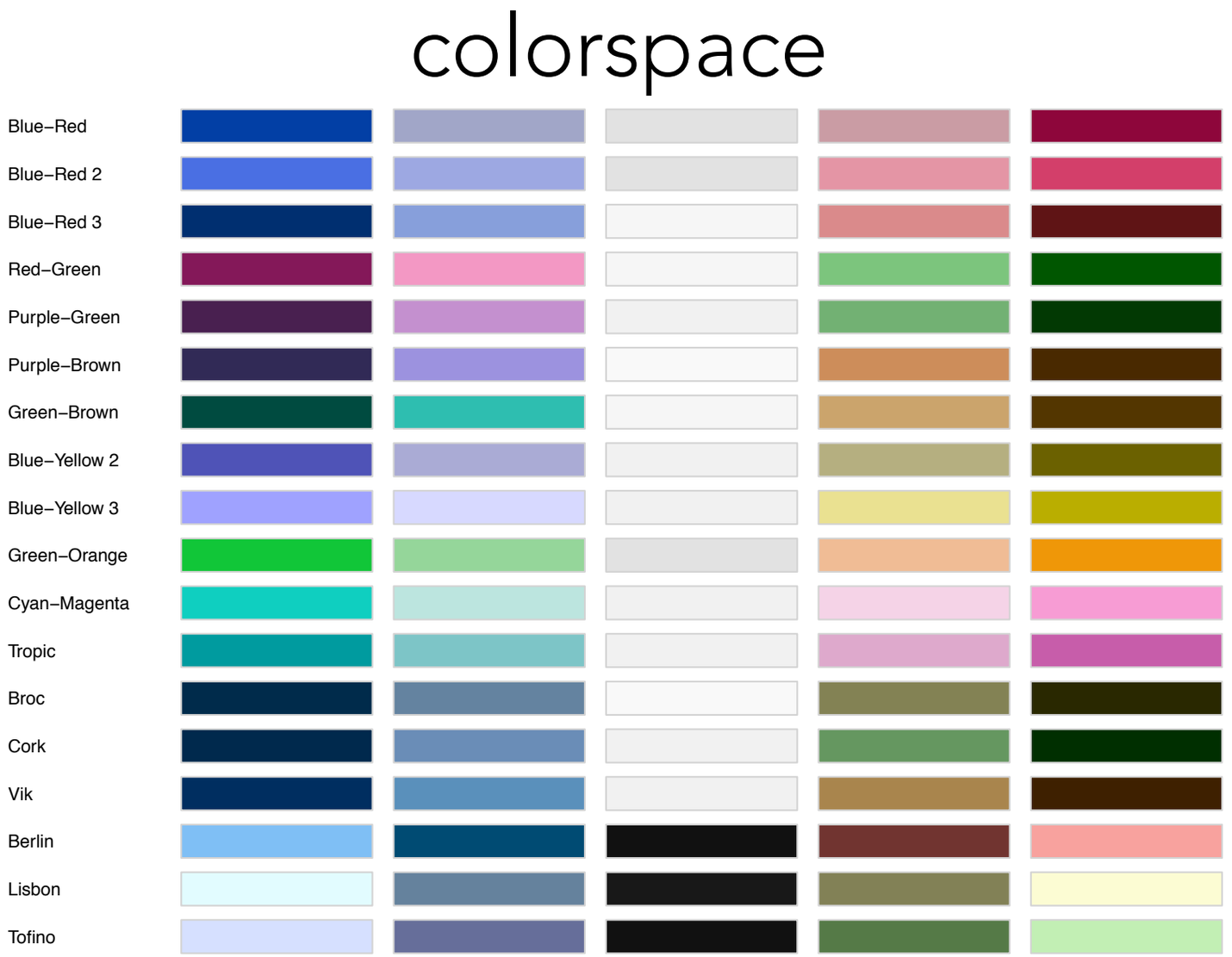
DIVERGING COLOR PALETTES

rcartocolor




```
Ordered categories: scale_color_carto_d(palette = "Tropic")
Continuous values: scale_color_carto_c(palette = "Tropic")
```

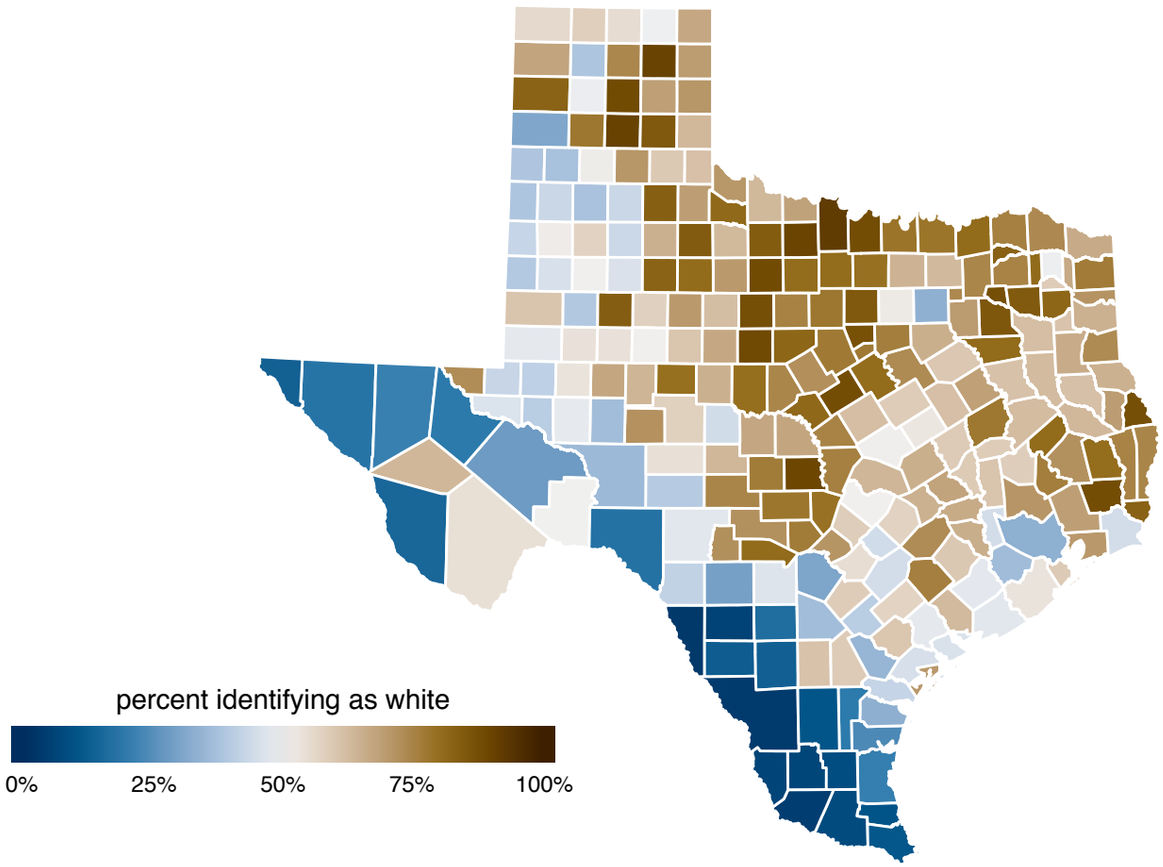
DIVERGING COLOR PALETTES




```
Ordered categories: scale_color_discrete_diverging(palette = "Vik")
Continuous values: scale_color_continuous_diverging(palette = "Vik")
```

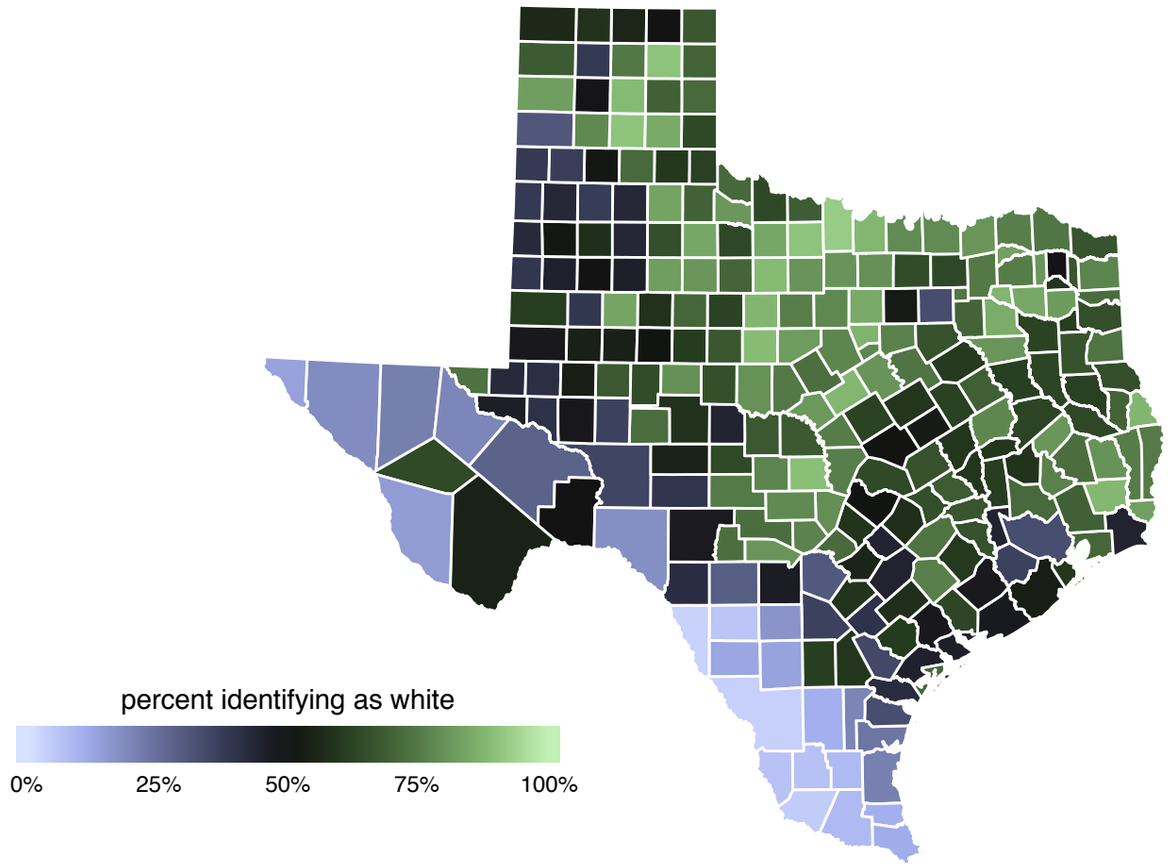
DIVERGING COLOR PALETTES

Light midpoint



Vik from colorspace

Dark midpoint



Tofino from colorspace

COMMON PITFALLS OF COLOR USE

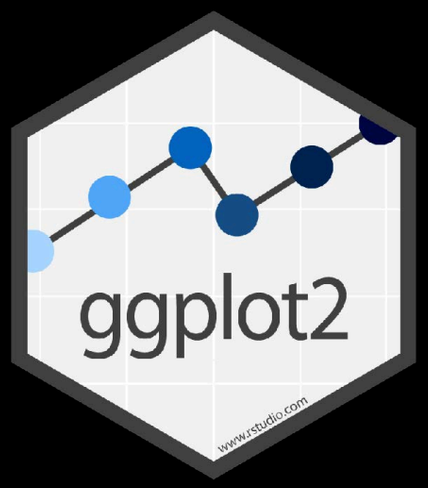
- Too many colors for discrete categories
- Using coloring without clear purpose
- Color gradients with perceptual discontinuities (e.g., rainbows)
- Not considering conversion to grayscale and color-vision deficiency

SAVING YOUR PLOTS

TWO WAYS I RECOMMEND



Let knitr create the plots for you



`ggsave()` function

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- This should be familiar to you already.
- Figure height and width can be specified in chunk options.

fig.height

fig.width

````
179
180 ```{r, fig.height = 6, fig.width = 10}
181 ggplot(filter(babynames, name == "Lionel" & sex == "M"),
182        aes(x = year, y = prop)) +
183   geom_ribbon(aes(ymax = prop, ymin = 0), alpha = 0.5) +
184   geom_line() +
185   annotate(geom = "rect", xmin = 1950, xmax = 1960, ymin = 0, ymax = Inf, fill = "red", alpha = 0.25) +
186   annotate(geom = "text", x = 1949, y = 3e-4, label = "Some text\nMore text", color = "red", hjust = 1) +
187   labs(title = "Popularity of the name Lionel")
188 ```
189
````

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- Where are the image files?

babynames-solutions.html

view-source:file://localhost/Users/czx412/OneDrive%20-%20University%20of%20Texas%20at%20San%20Antonio/Teaching/Data%20Visualization/activities/ant6973...

```
<div id="problem-8" class="section level4">
<h4><strong>Problem 8:</strong></h4>
<p>Finally, let's practice adding annotations to plots. An annotation is graphical element, like a point, line, box, or text label, that is added to a plot but is not part of your data set.
Annotations can be added using the <code>annotate()</code> function.</p>
<p>We're going to plot the popularity over time of the name Lionel for boys, and annotate some of interesting features with both rectangles and text. Here's a template:</p>
<pre class="r"><code>ggplot(filter(babynames, name == &quot;Lionel&quot; &amp; sex == &quot;M&quot;),
       aes(x = year, y = prop)) +
  geom_ribbon(aes(ymax = prop, ymin = 0), alpha = 0.5) +
  geom_line() +
  annotate(geom = &quot;rect&quot;, xmin = 1950, xmax = 1960, ymin = 0, ymax = Inf, fill = &quot;red&quot;, alpha = 0.25) +
  annotate(geom = &quot;text&quot;, x = 1949, y = 3e-4, label = &quot;Some text\nMore text&quot;, color = &quot;red&quot;, hjust = 1) +
  labs(title = &quot;Popularity of the name Lionel&quot;)</code></pre>
<p><img
src="
xL+3gCAo9Q/bPrQvlQol2tQgKD60+INQ6Ium65k7M5lpurHeZe58853vnnvuuWfvBoi5qliWkROBFpgwIRgw4pOHi4g0K5CEh6AXBqFXUR0rXalMAjZPC3olW99Dwntf2dXd/p+tt0YdFSBxH2Kz5qgLiI8B8KdVy3YBevqRHz/qWh72Yui3MUDEL3g44WPXw3M+fol
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YpfoWYXG+fAPPI6tJnNwb7ClP7IyF+D+bjOtCpkhz6CFrIa/I6sFtNl8auFXGMTP34sNwI/JhkgEtmDz14ySfaRcTIBInmKPE32kxyyE2Tv+thKbEVePDfW/byMM1Kmm0XdObS7oGD/MypMXFPXrCwOtoYjyyn7BV29/MZfsVzpLDdRtuIZnbpXzvlf+ev8MvYr/Gqk
4H/kV/G3csdazLuyTMPsbFhzdlUabQbjFvDRmcWJxR3zcfHkVw9GfpbJmeev9F08WW8uDkaslwX6avlWGU6NRKz0g/SHtCy9J30o/ca9zX3Kfc19zn3BXQKRO8ud477hLnAfc1/G9mrzGlrfexZ5GLdn6ZZrrEohI2wVHhZywjbhUWEy8icMCGNCUdiBlq3r+xafL54
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WY2zezO9Pzq85mdnu7q6qpvz6y4zzxVjmuKUBBAAAEEEEAAAQQQQAABBBBAAAEEE8l4gkvcjYAAIIIAAAggggAACCCCAAAIIIAAAAlaAADBvBAQQQAABBBBAAAEEEAAAQQQQCAkAgSA
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QQQAABBBBAAAEEEAAAQQQQAABBEiQAA4JDeSYSCAAAIIIAAAggggAACCCAAAIIIAAAggggAACCCAAAIIIAAAggggAACCCAAAIIIAAAiERIAAckhvJMBBAAAEEEAAAQQQQAABBBBAAAEEEAAAQQQQAABBBBAAAIEiCBBQg
ZUrV8q0adPKxYhEIlKvXj1p0KCBbLnlllK3btlyzymkClOnTpVVqlbZIXfp0kWaN2+eU8OfOXOmLFu2zPapQ4cO0rpl65zqX012Zu7cuTJixAiZPXu2bNy4Udq3by8t0JsmrsPE1suR5p1m5KSkjLD2WabbUSDQVUpqcz
U8eWXX7ZNDhs2TM4777yqNM85CMh9990nF154oZVo06aNLFy4sFoqFfl9UK2L5MHJuf57Ow8I6SICCCCAAAIIIAAAggggAACCCAAAII5JgAU0Dn2A3Jp+7odwe++uorG9DSbOBHH300n7pPX5MITJw4Uf74xz/a4GiSw6HY9f3338tzZ52lWfBXB7fzzjtXeoyFYF
zpFE5AAAEEEEAAAQQQQAABBBBAAAEEEEAAAQQQQACBWhNgCuhao8/tC++1114SjUY366QGfdevXy+rV6+2U+du2LDB1lm3bp389a9/lW7duknv3r03O48duS+g2YS77LKLnQ65fv36ud/kVvvld6euZCMasiNachgAACCCCAA
AIIIAAAggggAACCCAAAIIIAAAggggAACCCAAAII5JgAU0Dn2A3Jp+7odwe++uorG9DSbOBHH300n7pPX5MITJw4Uf74xz/a4GiSw6HY9f3338tzZ52lWfBXB7fzzjtXeoyFYF
ZpFE5AAAEEEEAAAQQQQAABBBBAAAEEEEAAAQQQQACBWhNgCuhao8/tC++1114SjUY366QGfdevXy+rV6+2U+du2LDB1lm3bp389a9/lW7duknv3r03O48duS+g2YS77LKLnQ65fv36ud/kVvvld6euZCMasiNachgAACCCCAA
AIIIAAAggggAACCCAAAII5JgAU0Dn2A3Jp+7odwe++uorG9DSbOBHH300n7pPX5MITJw4Uf74xz/a4GiSw6HY9f3338tzZ52lWfBXB7fzzjtXeoyFYF
z
```

KNITR



- By default, knitr creates standalone HTML documents by encoding the image and embedding it inline with the document.

- That means that there is no separate image file sitting somewhere on your computer.

- This makes the HTML file portable!

- But if you need to *save* one of the images as a file, you can do so easily from your web browser.

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In-class Activity: Changing Ba

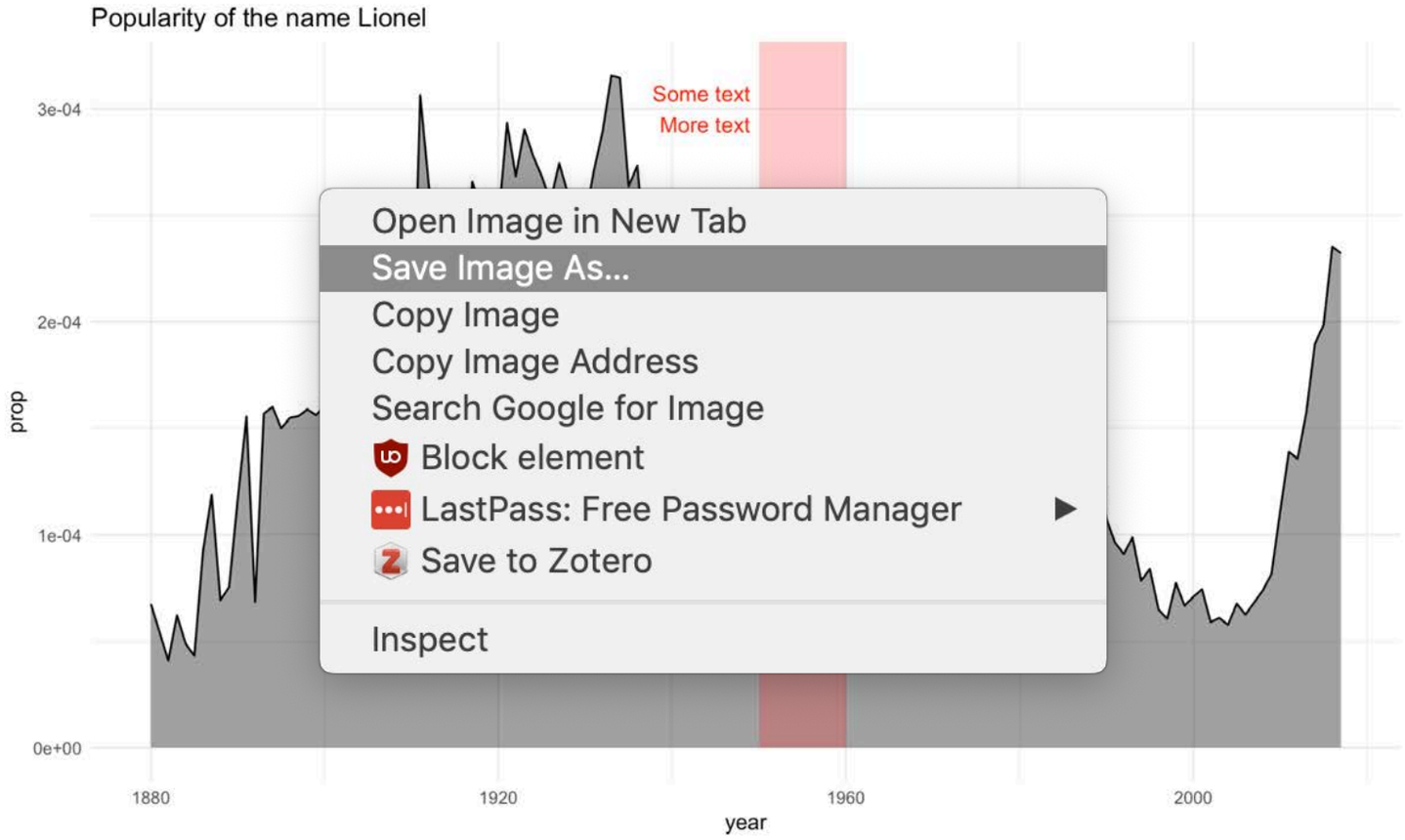
File | file://localhost/Users/czx412/OneDrive%20-%20University%20of%20Texas%20at%20San%20Antonio/Teaching/Data%20Visualization/ac...

Problem 8:

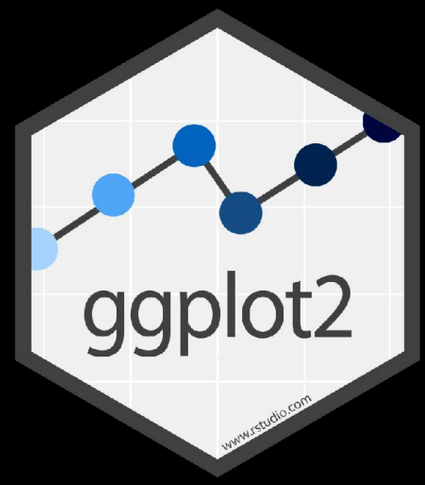
Finally, let's practice adding annotations to plots. An annotation is graphical element, like a point, line, box, or text label, that is added to a plot but is not part of your data set. Annotations can be added using the `annotate()` function.

We're going to plot the popularity over time of the name Lionel for boys, and annotate some of interesting features with both rectangles and text. Here's a template:

```
ggplot(filter(babynames, name == "Lionel" & sex == "M"),
       aes(x = year, y = prop)) +
  geom_ribbon(aes(ymax = prop, ymin = 0), alpha = 0.5) +
  geom_line() +
  annotate(geom = "rect", xmin = 1950, xmax = 1960, ymin = 0, ymax = Inf, fill = "red", alpha = 0.25) +
  annotate(geom = "text", x = 1949, y = 3e-4, label = "Some text\nMore text", color = "red", hjust = 1) +
  labs(title = "Popularity of the name Lionel")
```



GGSAVE



- Save a plot as an image file

```
ggsave(<filename>, <plot>, ...)
```

function

Name of file you want to create

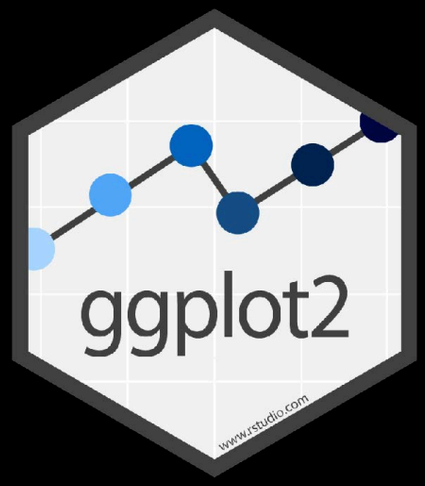
Name of plot object in R (if any)

Options including width, height, and units

You provide the path and file extension (e.g., ".png" or ".pdf")

If nothing is specified here, it'll use the last plot that you created!

GGSAVE



- Example

```
ggplot(starwars, aes(x = eye_color)) +
  geom_bar()

ggsave("starwars-eye-color.pdf", width = 10, height = 5)
```

No path provided, so file will be saved in working directory

Function knows to create a PDF file

GGSAVE

- Example

```
ggplot(starwars, aes(x = eye_color)) +
  geom_bar()

ggsave("07_viz-spatial/starwars-eye-color.pdf",
       width = 10, height = 5)
```

Provide a path to save file in subfolder of project home (note: folder must exist)

GGSAVE

- Example

```
ggplot(starwars, aes(x = eye_color)) +
  geom_bar()

ggsave("~/Documents/Teaching/DataViz/activities/07_viz-
spatial/starwars-eye-color.pdf", width = 10, height = 5)
```

An absolute path will work but it is usually a sign of potential problems. It suggests you're not using a project-based workflow, and it will not work on other computers.

HOW DO I GET THE SIZE RIGHT?

- You can eventually develop an intuition about sizing image files…
- … but it still involves some trial and error.
- In R Studio, you can experiment using Export > Save as PDF… in the Plots pane.

HOW DO I GET THE SIZE RIGHT?

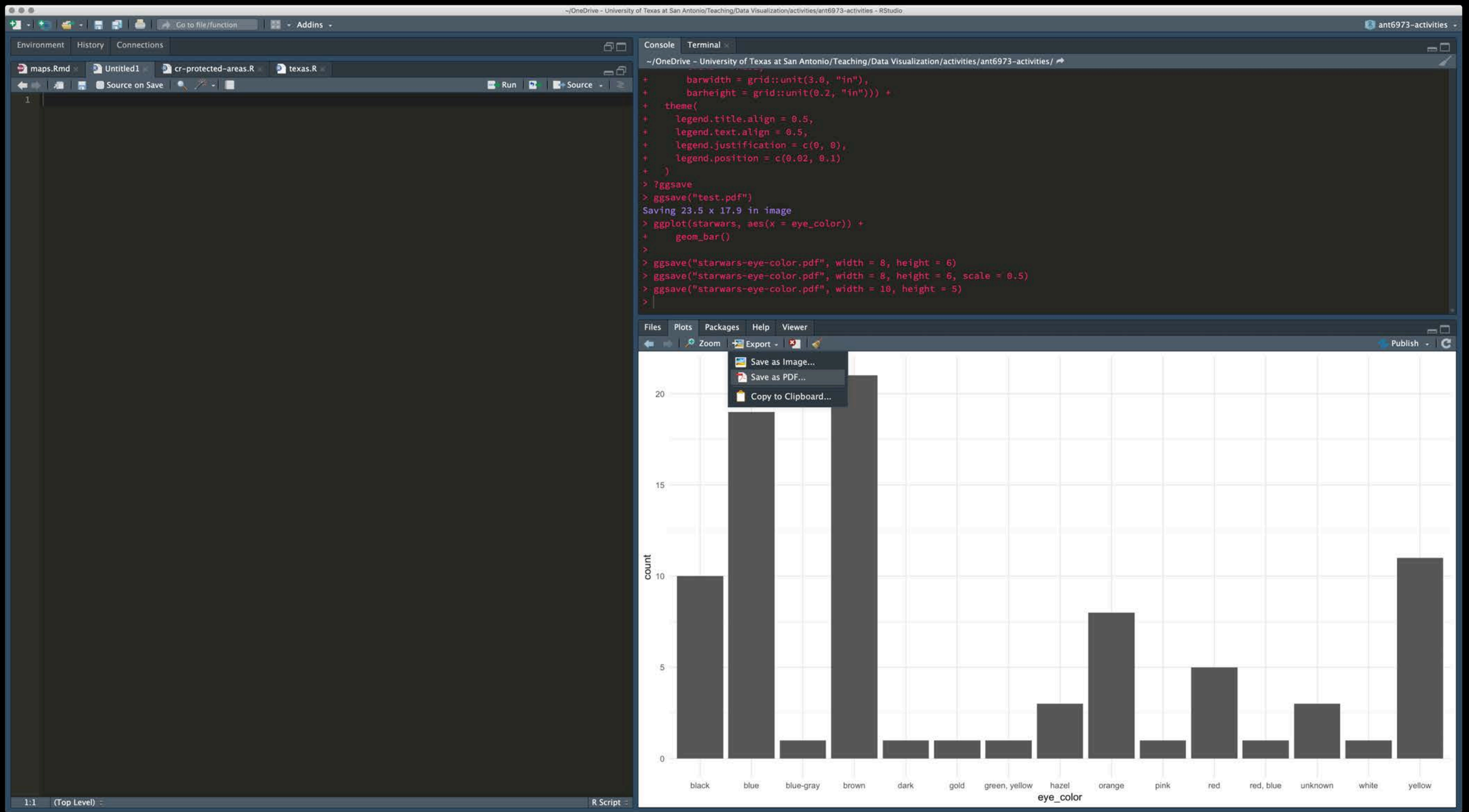


IMAGE FORMATS

- Refer to Wikle Ch. 27, "Understanding the most commonly used image file formats"

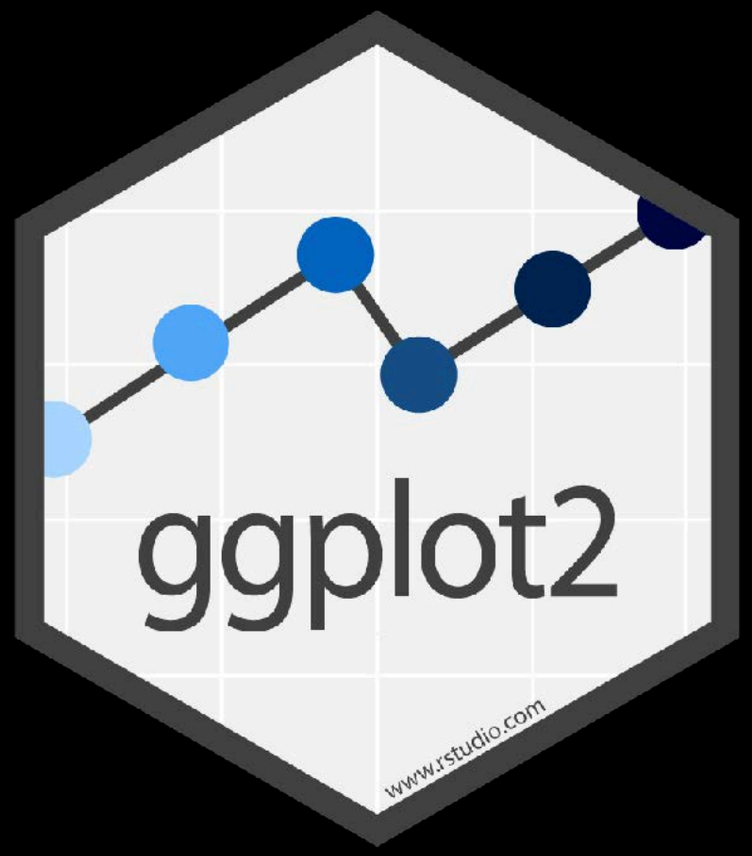
- Rules of thumb:

    1. Use PDF whenever possible, especially for creating high-quality, publication-ready files.

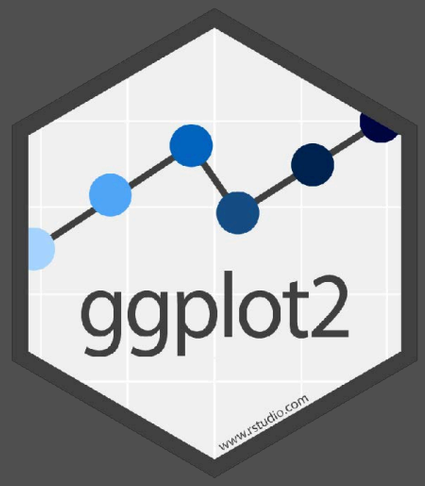
    2. Use PNG or TIFF for reports and web content, if required by journal, or for plots with huge amounts of data (e.g., millions of points).

    3. Use JPG only for photographs or if PNG file is too large.

VISUALIZING GEOSPATIAL DATA



ACTIVITY: MAPS



- Go to this week's assignments on the course website.
- Follow the instructions in the maps.Rmd file to learn some map-making skills.
- Submit the html report (not the Rmd) before next class.

